

MP2 Telling a Visual Narrative of your Data

FIFA World Cup Data Analysis Dataset: FIFA World Cup Summary (1930–2022)

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Context: As FIFA's data analyst, we explore historical World Cup trends to provide actionable insights for the 2026 World Cup taking place in the United States, Canada, and Mexico

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In [2]: #Importing libraries
import pandas as pd
import matplotlib.pyplot as plt
```

```
In [14]: #Loading the dataset
df = pd.read_csv('FIFAWorldCupSummary.csv')

#Data cleaning
print('df.shape:', df.shape)
# Checking for missing values
print(df.isnull().sum())
# Check data types
print(df.dtypes)

# Drop any rows with missing values (if any)
df = df.dropna()
print(df.shape)
```

```
df.shape: (22, 9)
YEAR                0
HOST                0
CHAMPION            0
RUNNER UP          0
THIRD PLACE        0
TEAMS              0
MATCHES PLAYED     0
GOALS SCORED       0
AVG GOALS PER GAME 0
dtype: int64
YEAR                int64
HOST                object
CHAMPION            object
RUNNER UP          object
THIRD PLACE        object
TEAMS              int64
MATCHES PLAYED     int64
GOALS SCORED       int64
AVG GOALS PER GAME float64
dtype: object
(22, 9)
```

Fig 1:Championship Wins by Country: Which countries have won the most World Cups? Here we count total wins per country, filtering to only those with 2 or more titles to keep the chart readable.

```
In [ ]: # Figure 1: World Cup Championship Wins
champion_counts = df['CHAMPION'].value_counts()
champion_counts = champion_counts[champion_counts > 1]

colors = ['gold', 'steelblue', 'tomato', 'mediumseagreen', 'mediumpurple']

fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(10, 4))

ax1.bar(champion_counts.index, champion_counts.values, color=colors)
ax1.set_title('Wins by Country')
ax1.set_ylabel('Wins')

ax2.pie(champion_counts.values, labels=champion_counts.index, colors=colors,
ax2.set_title('Share of Titles')

plt.suptitle('Fig 1: World Cup Championship Wins', fontweight='bold')
plt.tight_layout()
plt.show()
```

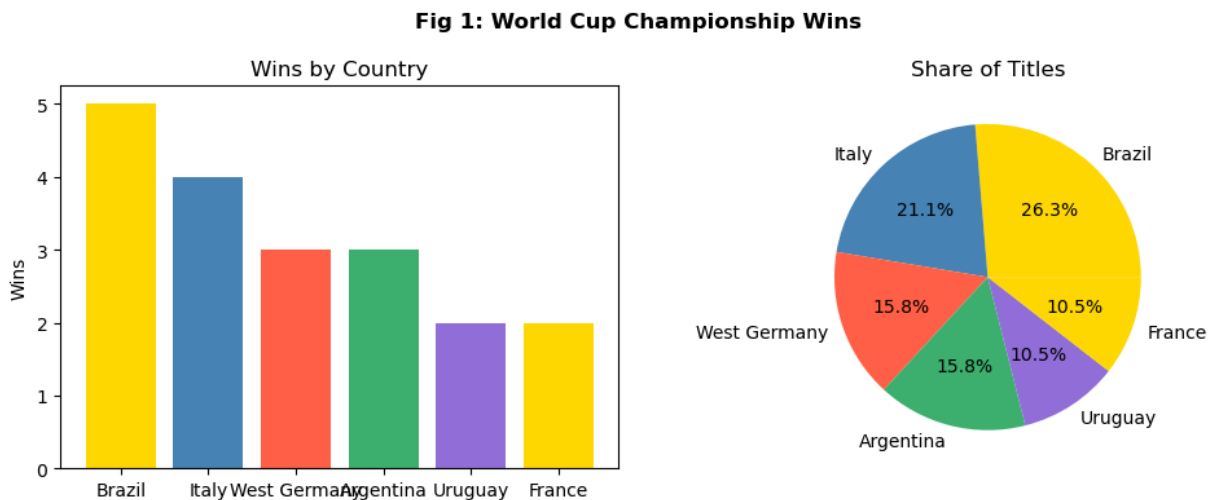


Fig 1: Total World Cup wins by country, showing Brazil as the most dominant nation with 5 championships.

Fig 2: Average Goals Per Game Over Time: Has the World Cup become more or less exciting over the years? This line chart tracks the average goals scored per game across every tournament since 1930.

```
In [ ]: # Figure 2: Average Goals Per Game Over Time
fig, ax = plt.subplots(figsize=(10, 4))

ax.plot(df['YEAR'], df['AVG GOALS PER GAME'], color='steelblue', marker='o')
ax.set_title('Avg Goals Per Game Over Time')
ax.set_xlabel('Year')
ax.set_ylabel('Avg Goals Per Game')

plt.suptitle('Fig 2: Scoring Trends Across World Cups', fontweight='bold')
plt.tight_layout()
plt.show()
```

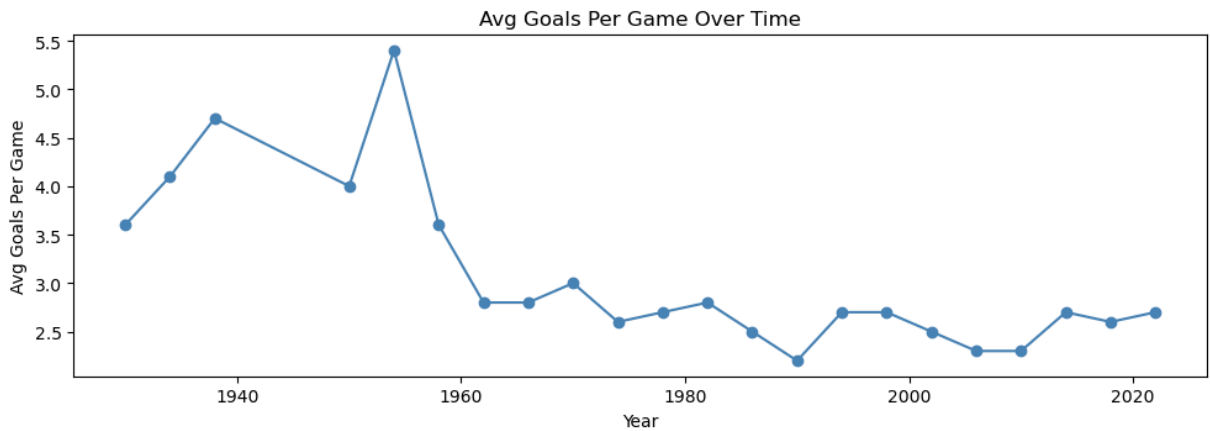
Fig 2: Scoring Trends Across World Cups

Fig 2: Average goals per game across every World Cup, showing a steady decline in scoring since the high of 5.4 in 1954.

Fig 3: Tournament Growth Over Time: How has the World Cup grown as a tournament? This line chart tracks the number of teams and matches played across every World Cup since 1930.

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In [ ]: #Figure 3: Growth of Teams and Matches Over Time
fig, ax = plt.subplots(figsize=(10, 4))

ax.plot(df['YEAR'], df['TEAMS'], color='steelblue', marker='o', label='Teams')
ax.plot(df['YEAR'], df['MATCHES PLAYED'], color='tomato', marker='o', label='Matches Played')
ax.set_title('Teams and Matches Over Time')
ax.set_xlabel('Year')
ax.set_ylabel('Count')
ax.legend()

plt.suptitle('Fig 3: World Cup Tournament Growth', fontweight='bold')
plt.tight_layout()
plt.show()
```

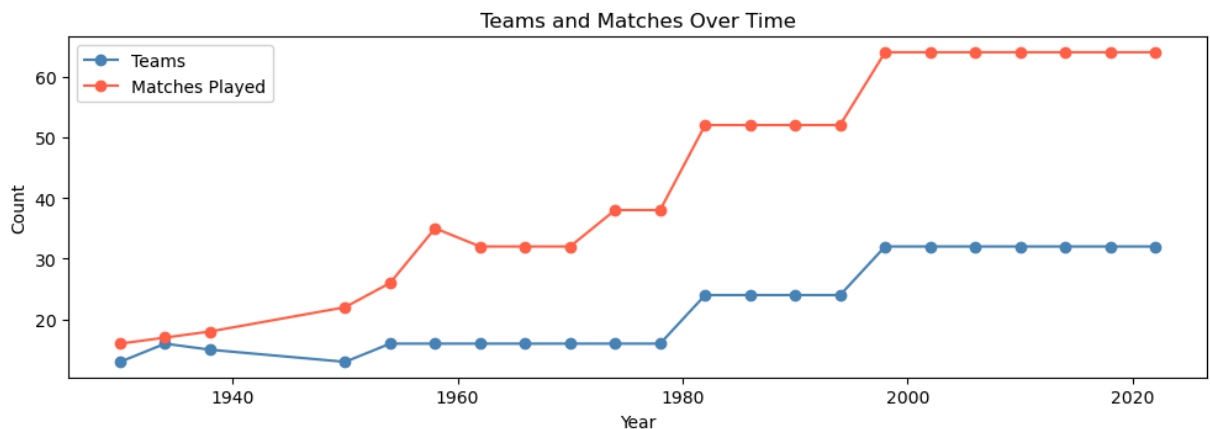
Fig 3: World Cup Tournament Growth

Fig 3: Growth of the World Cup tournament over time, showing significant jumps in teams and matches in 1982 and 1998.

Fig 4: Top 10 Countries by Podium Finishes: Which countries have been the most consistently dominant? Here we count total top 3 finishes (1st, 2nd, and 3rd place) for each country.

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In [ ]: # Figure 4: Overall World Cup Dominance
podium = pd.concat([df['CHAMPION'], df['RUNNER UP'], df['THIRD PLACE']])

podium_counts = podium.value_counts().head(10)

colors = ['gold', 'steelblue', 'tomato', 'mediumseagreen', 'mediumpurple',
          'orange', 'pink', 'brown', 'gray', 'cyan']

fig, ax = plt.subplots(figsize=(10, 5))

ax.barh(podium_counts.index[::-1], podium_counts.values[::-1], color=colors)
ax.set_title('Top 10 Countries by Podium Finishes')
ax.set_xlabel('Total Podium Finishes')

plt.suptitle('Fig 4: Overall World Cup Dominance', fontweight='bold')
plt.tight_layout()
plt.show()
```

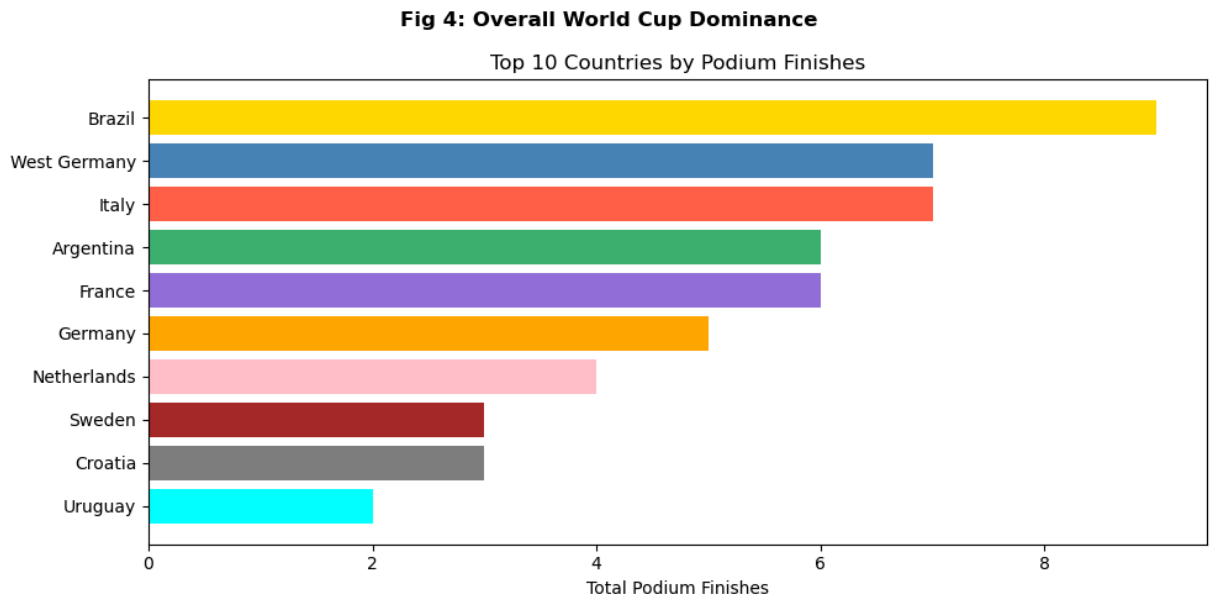


Fig 4: Top 10 countries by total podium finishes, highlighting Germany and Brazil as the most consistently dominant nations.

Fig 5: Host Country Performance: Does hosting the World Cup give a country an advantage? Here we check whether each host country finished in 1st, 2nd, 3rd place, or did not place at all.

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In [ ]: #Figure 5: Correlation Between Teams and Goals

def host_result(row):
    if row['HOST'] == row['CHAMPION']:
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        return '1st Place'
    elif row['HOST'] == row['RUNNER UP']:
        return '2nd Place'
    elif row['HOST'] == row['THIRD PLACE']:
        return '3rd Place'
    else:
        return 'Did Not Place'

df['HOST RESULT'] = df.apply(host_result, axis=1)

result_counts = df['HOST RESULT'].value_counts()

colors = ['tomato', 'steelblue', 'gold', 'mediumseagreen']

fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(10, 4))

ax1.bar(result_counts.index, result_counts.values, color=colors)
ax1.set_title('Host Country Results')
ax1.set_ylabel('Count')

ax2.pie(result_counts.values, labels=result_counts.index, colors=colors, autopct='%1.1f%%')
ax2.set_title('Share of Host Results')

plt.suptitle('Fig 5: Did Hosting Give an Advantage?', fontweight='bold')
plt.tight_layout()
plt.show()

```

Fig 5: Did Hosting Give an Advantage?

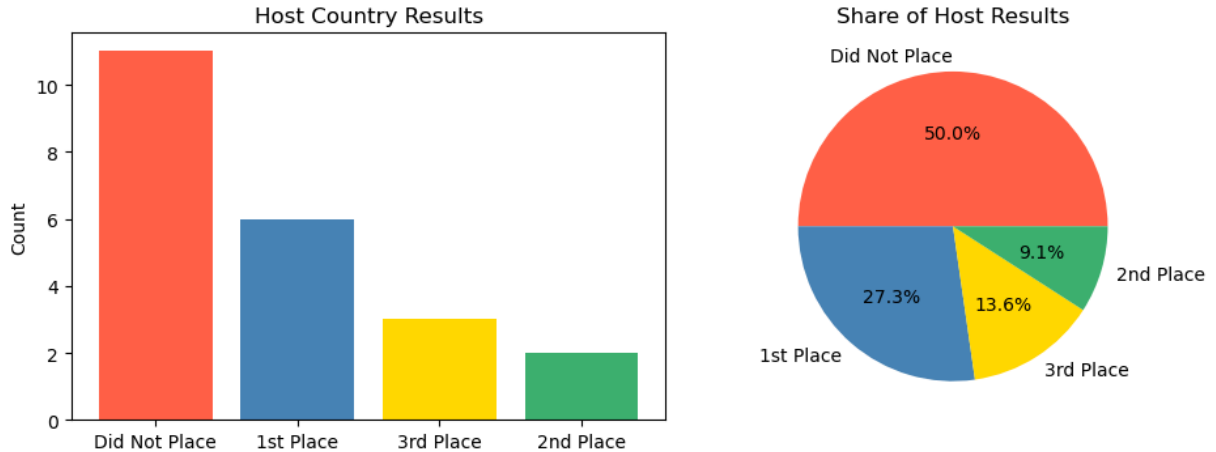


Fig 5: Host country performance across all tournaments, showing that while most hosts did not place, a notable number won or finished on the podium.

Actionable Advice:

Based on my analysis of the FIFA World Cup data from 1930 to 2022, I have three recommendations. First, FIFA should continue expanding the tournament, the data clearly shows that every time we increased the number of teams (1982 and 1998), the number of matches and total goals scored went up significantly, meaning more games and more excitement for fans. Second, since average goals per game have steadily declined since 1954, I recommend FIFA look into rule changes or format adjustments

that could encourage more attacking play and bring scoring back up. However, this could also be a result of defensive play improving significantly. Third, I recommend we start collecting attendance and viewership data for each match. Right now the dataset only tells us what happened on the field, but if we had audience data we could identify which matchups and which rounds drive the most interest, helping us make smarter decisions around scheduling and marketing for future tournaments.

With the 2026 World Cup coming to North America, hosted across the United States, Canada, and Mexico, this will be the largest World Cup in history, featuring 48 teams for the first time. This is a massive opportunity for FIFA, and the trends in our data back it up. Expanding to 48 teams means more matches, more goals, and a bigger global audience than ever before. North America is one of the largest sports markets in the world, and bringing the tournament here gives us a chance to grow the sport in a region where soccer is still gaining popularity.